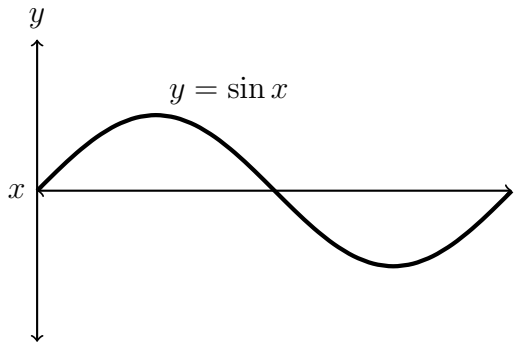
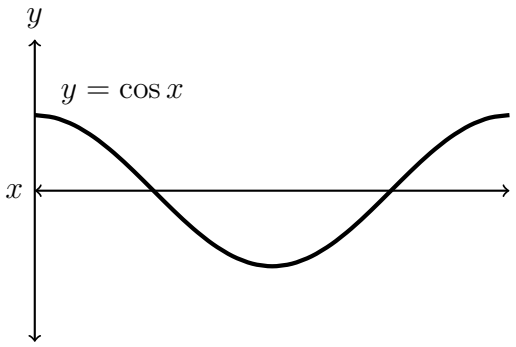
MATH 195: 11/2
WAVES

Graphs of $\sin x$ and $\cos x$.

Both $\sin x$ and $\cos x$ have a period of 2π, a maxium of 1, and a minimum of -1. The graphs are horizontal translations of each other. You can think of them as differing where in the wave they start.



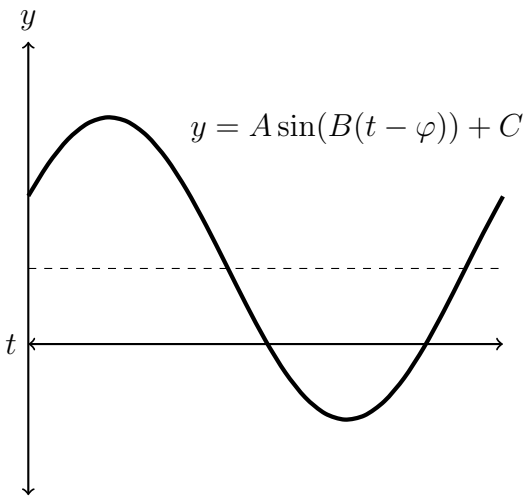


Waves.

sin and cos can be used to model waves, using geometric transformations to modify the basic picture.

$$y = A\sin\big(B(t-\varphi)\big) + C$$

- The *amplitude* $|A|$ is the height from the center to top of the wave.
- The *period* $2\pi/B$ is how long it takes the wave to repeat.
- The *vertical shift* C is the height the wave is centered on.
- φ ("phi") gives the *phase shift*, where in the wave it starts at time 0. Alternatively, you can get the phase shift for starting at the top, bottom, or middle of the wave by using $\pm\sin$ or $\pm\cos$ without φ.



Understanding geometric transformations and waves

Use a graphing calculator such as the one at `desmos.com` for these exercises. The key thing to keep in mind is geometric transformations. Try to answer these based on your understanding before you check your answer with the graphing calculator.

(1) If a function $f(x)$ achieves a maximum output of 1, what is the maximum of $3f(x)$? Confirm your answer by using a graphing calculator to graph $3\sin x$.
(2) If a function $g(x)$ achieves a minimum output of -1, what is the maximum of $-4g(x)$? Confirm your answer by using a graphing calculator to graph $-4\cos x$.
(3) In general, if a function $h(x)$ has a range of $[-1, 1]$, what is the range of $A \cdot h(x)$, where A is a nonzero constant? Check your answer using a graphing calculator.
(4) If a function $j(x)$ has a domain of $[-1, 1]$, what domain does $h(2x)$ have? Confirm your answer by using a graphing calculator to graph $\arcsin(2x)$.
(5) A *sawtooth wave* $s(x)$ can be plotted in desmos by plotting `mod(x,1)`. It has a period of 1, as you can confirm by looking at its graph. What are the periods of $s(x/2)$ and $s(4x)$? Confirm your answers by plotting `mod(x/2,1)` and `mod(4x,1)`.
(6) If a function $k(x)$ has a period of 2π, what are the periods of $k(\pi x)$ and $k(x/3)$? Confirm your answer by graphing $\sin(\pi x)$ and $\sin(x/3)$.
(7) In general, if a function $\ell(x)$ has a period of 2π, what is the period of $\ell(Bx)$, where B is a nonzero constant. Check your answer using a graphing calcultor.
(8) The function $\cot x$ has a period of π, as you can confirm by looking at its graph. What is the period of $\cot(Bx)$, where B is a nonzero constant? Check yoru answer using a graphing calculator.
(9) If a function $m(x)$ has a domain of $[-1, 1]$, what is the domain of $m(x - 1)$ Check your answer by using a graphing calculator to graph $\arccos(x - 1)$. More generally, what is the domain of $m(x - D)$, where D is a constant?
(10) If a function $n(x)$ has a range of $[-1, 1]$, what is the range of $n(x) + 3$? Check your answer by using a graphing calculator to graph $\sin(x) + 3$. More generally, what is the domain of $n(x) + C$, where C is a constant?
(11) If a function $p(x)$ has a range of $[-1, 1]$, what is the range of $2p(x) + 3$? Check your answer by using a graphing calculator to graph $2\sin(x) + 3$.
(12) In general, what is the range of $Ap(x) + C$, where $A \neq 0$ and C are constants?
(13) Put it all together. What are the range and period of $A\sin(Bx) + C$ where $A, B \neq 0$ and C are constants? Confirm your answer using a graphing calculator.

Graphs of waves

Use a graphing calculator to assist you in checking these.

(1) Determine the amplitude, minimum, maximum, and period of
$$w(t) = -2\sin(t/2) + 1.$$
(2) Determine the amplitude, minimum, maximum, and period of
$$v(x) = 3\cos(4x) - 3.$$
(3) Determine the period of
$$u(x) = \sin(\pi x/180).$$
What does this tell you about radians versus degrees and the graphs of trig functions?